



GIRIJANANDA CHOWDHURY UNIVERSITY

Hathkhowapara, Azara, Guwahati 781017, Assam

School of Engineering & Technology

DEPARTMENT OF CIVIL ENGINEERING

B.Tech - Civil Engineering

Semester V

Sl. No	Course Code	Course Name	Hours per week			Credits
			L	T	P	
1.	BCE23501T	Structural Design -I	3	1	0	4
2.	BCE23502T	Environmental Engineering	3	0	0	3
3.	BCE23512P	Environmental Engineering Lab	0	0	2	1
4.	BCE23503T	Engineering Economics, Estimation & Costing	3	0	0	3
5.	BCE23504T	Hydrology & Water Resources Engineering	3	0	0	3
6.	BCE235PE10T	Plumbing (Water and Sanitation)	2	0	0	2
7.	BCE235PE10P	Plumbing (Water and Sanitation) Lab	0	0	2	1
8.	MOPEC-1	<i>Multidisciplinary Open Electives Courses -1</i> Environmental Studies	3	0	0	3
9.	MNC-AU-05	Professional Practice, Law & Ethics	3	0	0	0
10.	BCE-SI-02	Summer Internship-2*	0	0	4	2
		Total	20	1	8	22
11.		Minor/Honours/Value Added Courses (Optional)	3	0	0	3

* summer internship during end of semester 4 to be evaluated in semester 5



BCE23501T	STRUCTURAL DESIGN - I	L	T	P	C
		3	1	0	4
Pre-requisite: Nil					
Course Objectives:					
The objective of structural design is to introduce the theory and application of analysis and design of reinforced concrete structures. The course focuses on understanding the behavior of reinforced concrete components and systems subjected to gravity as well as lateral loads.					
Expected Course Outcome:					
Upon completion of this course, the student will be able to					
1. Understand material properties and behavior of reinforced concrete.					
2. Understanding Design Philosophies and Principles.					
3. Design and analysis of RC elements under different loading conditions, including flexure, shear, and torsion.					
4. Application of relevant building codes (e.g., IS 456, SP 16, SP 34, IS:13920) in the analysis and Design of RC Elements.					
Module:1 Introduction to reinforced concrete					10 hours
Concept of reinforced concrete, stress strain characteristics of concrete and steel reinforcement, elastic theory, singly reinforced, balanced section, under reinforced section and over reinforced section. Design methodologies; Introduction to working stress method; Limit state method of design: Concepts of probability and reliability, characteristic loads, characteristic strength, partial safety factors for loads and materials, introduction to limit states of collapse in flexure, direct compression, shear and limit states of serviceability in deflection and cracking.					
Module:2 Design of Beams					6 hours
Design and detailing of Beams for flexure, shear, torsion, and bond: singly and doubly reinforced, rectangular and flanged; Continuous Beams; Serviceability criteria: deflection.					
Module:3 Design of Slabs					6 hours
Design and detailing of Slabs: one way and two way, simply supported and continuous.					
Module:4 Design of Columns					5 hours
Design and detailing of Columns: Short Columns under axial compression, axial compression with uniaxial bending, and axial compression with biaxial bending; Slender Columns.					
Module:5 Design of Footings					5 hours
Design and detailing of Footing: isolated and combined.					
Module:6 Design of Staircases					4 hours
Introduction to design and detailing of Staircase; Design of Doglegged, Open well type staircases					
Module:7 Introduction to IS 13920-2016					4 hours
Introduction to IS 13920-2016: Ductile detailing of reinforced concrete structures subjected to seismic forces - Code of practice. [CED 39: Earthquake Engineering]					
Total Lecture hours					40 hours
Text Book(s)					
1.	Reinforced Concrete Design, S U Pillai & Devdas Menon. Tata McGraw Hill.				
Reference Books					
1.	IS:456 -2000 -Plain and Reinforced concrete - Code of practice, BIS India.				
2.	IS 875 (1987), Part I- Design Loads (Other than earthquake) for Buildings and Structures (Dead Loads). IS 875 (1987), Part II- Design Loads (Other than earthquake) for Buildings and Structures (Imposed Loads). IS 875 (2015), Part III- Design Loads (Other than earthquake) for Buildings and Structures (Wind Loads). IS 875 (1987), Part IV- Design Loads (Other than earthquake) for Buildings and Structures (Snow Loads).				
3.	Design of Reinforced Concrete Structures (Limit State) – A. K. Jain, 1 st Edition, Nemchand Brothers, Roorkee.				
4.	P.C. Verghese, Limit State Design of Reinforced Concrete, PHI				
5.	Design of Reinforced Concrete Structures, N. Subramanian, OUP India				
Mode of Assessment: Continuous Assessment Test, Quizzes, Assignments, Final Assessment Test					



BCE23502T	ENVIRONMENTAL ENGINEERING	L	T	P	C
		3	0	0	3
Pre-requisite: Nil					
Course Objectives:					
The objective is to equip students with the essential knowledge and skills to identify, assess, and propose sustainable solutions for critical environmental challenges related to water supply and treatment, wastewater management, air and noise pollution control, and solid waste disposal, including an overview of practical building plumbing applications					
Expected Course Outcome:					
Upon completion of this course, the student will be able to					
1. Understand the impact of humans on environment and environment on humans					
2. Evaluate and value the effect of the pollutants on the environment.					
3. Be able to develop and propose comprehensive strategies for the control, reduction, and monitoring of various forms of pollution					
4. Be able to select appropriate techniques for the treatment of water, wastewater, solid waste, and contaminated air, demonstrating an understanding of fundamental environmental legislation.					
Module:1Water					10 hours
Sources of Water and quality issues, water quality requirement for different beneficial uses, Water quality standards, water quality indices, Water Supply systems, Need for planned water supply schemes, Components of water supply system; Distribution system, service reservoirs and design.					
Water Treatment: aeration, sedimentation, coagulation flocculation, filtration, disinfection, advanced treatments like adsorption, ion exchange, membrane processes					
Module:2 Sewage					9 hours
Domestic and Storm water, Quantity of Sewage, Sewage flow variations. Conveyance of sewage- Sewers shapes design parameters, operation and maintenance of sewers, Sewer appurtenances, Design of sewerage systems. Storm Water- Quantification and design of Storm water; Sewage and Sullage, Wastewater treatment, aerobic and anaerobic treatment systems, suspended and attached growth systems.					
Module:3 Air and Noise					8 hours
Composition and properties of air, Quantification of air pollutants, Monitoring of air pollutants, Air pollution- Occupational hazards, Urban air pollution automobile pollution, Air quality standards, Control measures for Air pollution,					
Noise- Basic concept, measurement and various control methods.					
Module:4 Solid waste management					6 hours
Municipal solid waste, Composition and various chemical and physical parameters of MSW, MSW management: Collection, transport, treatment and disposal of MSW. Solid waste from construction activities, biomedical wastes. Disposal of solid waste - segregation, reduction at source, recovery and recycle. Disposal methods- Integrated solid waste management. Solid waste management and monitoring/control of environmental pollution.					
Module:5 Building Plumbing					3 hours
Introduction to various types of home plumbing systems for water supply and waste water disposal, high rise building plumbing, Pressure reducing valves, Break pressure tanks, Storage tanks, Building drainage for high rise buildings, various kinds of fixtures and fittings used.					
Total Lecture hours					36 hours
Text Book(s)					
1.	Environmental Engineering, M.P. Poonia, SC. Sharma, Santosh Kumar, Khanna Book Publishing Co., New Delhi.				
2.	Air Pollution and Control, Keshav Kant, Rajni Kant, Khanna Book Publishing Co., New Delhi.				
Reference Books					
1.	Wastewater Engineering, Treatment, Disposal and Reuse, MetCalf and Eddy, Tata McGraw-Hill, New Delhi.				
2.	Elements of Water Pollution Control Engineering, O.P. Gupta, Khanna Publishing House, New Delhi				
3.	Plumbing Engineering. Theory, Design and Practice, S.M. Patil, 1999				
4.	Manual on Water Supply and Treatment. Ministry of Urban Development, New Delhi.				
5.	Elements of Environmental Pollution Control Engineering, O.P. Gupta, Khanna Publishing House, New Delhi				
Mode of Assessment: Continuous Assessment Test, Quizzes, Assignments, Final Assessment Test					



BCE23503T	Engineering Economics, Estimation & Costing	L	T	P	C
		3	0	0	3
Pre-requisite: Nil					
Course Objectives:					
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	2. Mankiw Gregory N. (2002), Principles of Economics, Thompson Asia		
	3. V. Mote, S. Paul, G. Gupta(2004), Managerial Economics, Tata McGraw Hill		
	4. Misra, S.K. and Puri (2009), Indian Economy, Himalaya		
	5. Pareek Saroj (2003), Textbook of Business Economics, Sunrise Publishers		
	6. M Chakravarty, Estimating, Costing Specifications & Valuation		
	7. Joy P K, Handbook of Construction Management, Macmillan		
	8. B.S. Patil, Building & Engineering Contracts		
	9. Relevant Indian Standard Specifications.		
	10. World Bank Approved Contract Documents and FIDIC Contract Conditions.		
	12. Acts Related to Minimum Wages, Workmen’s Compensation, Contract, and Arbitration		
	13. Typical PWD Rate Analysis documents.		
Mode of Assessment: Continuous Assessment Test, Quizzes, Assignments, Final Assessment Test			
Recommended by Board of Studies			
Approved by Academic Council			Date



BCE23504T	HYDROLOGY AND WATER RESOURCES ENGINEERING	L	T	P	C
		3	0	0	3
Pre-requisite: Nil					
Course Objectives:					
1. To know the hydrological cycle and its different forms. 2. To obtain knowledge of groundwater and well hydrology. 3. To analyze hydrographs and Instantaneous Unit Hydrograph. 4. To understand the basics of water resources and cost allocation.					
Expected Course Outcome:					
Upon completion of this course, the student will be able to 1. To understand hydrological cycle, precipitation, evapotranspiration, infiltration and its measurements. 2. To apply basic knowledge of hydrology for calculating the runoff. 3. To analyze hydrographs and Instantaneous Unit Hydrograph. 4. To solve for water supply requirement along with cost allocations for the water resources development.					
Module:1 Introduction					3 hours
Hydrologic cycle, Schematic representation, System representation. Global Water Budget, History of Hydrology, Scope of Hydrology.					
Module:2 Precipitation					8 hours
Forms of precipitation, characteristics of precipitation in India, measurement of precipitation, rain gauge network, mean precipitation over an area, depth- area- duration relationships, maximum intensity/depth-duration-frequency relationship, Probable Maximum Precipitation (PMP), rainfall data in India. Forms and Types of Precipitation. Measurement of rainfall –Non-Recording and Recording Rain gauges, AWS, Radar measurement of rainfall, Selection of Rain gauge site, Rain gauge net-work, Adequacy of Rain Gauge Stations. Methods of computing average rainfall –Arithmetic mean method, Thiessen Polygon method, Isohyetal method. Interpolation and Adjustment of Missing Data –Station Year Method, Double mass curve. Related Numerical Problems. Evapotranspiration, measurement of evapotranspiration, potential evapotranspiration over India, actual evapotranspiration.					
Module:3 Infiltration and Run-off					6 hours
Introduction, Factors affecting infiltration, Infiltration capacity-Horton’s infiltration curve. Infiltration Indices, Numerical solutions, Measurement of infiltration–Flooding type Infiltrometer. Introduction and Estimation of runoff –Empirical formulae, Infiltration method, Hydrograph analysis, Rational method.					
Module:4 Hydrograph					8 hours
Introduction, Components of hydrograph, Base Flow Separation, Computation of direct Surface Runoff. Unit Hydrograph –Definition, Derivation of UH with assumptions, Limitations and uses, UH of different durations, Synthetic UH, Instantaneous UH, all related numerical problems.					
Module:5 Groundwater and Well Hydrology					6 hours
Forms of subsurface water, saturated formation, aquifer properties, geologic formations of aquifers, well hydraulics: steady state flow in wells, equilibrium equations for confined and unconfined aquifers, aquifer tests.					
Module:6 Introduction to Water Resources Engineering					3 hours
Fields of water resources engineering; problems of water resources engineering, economics in water resources engineering, Social aspects of water resources engineering, planning of water resources projects, the future of water resources engineering. Water resource in North East and its use.					
Module:7 Cost Allocation					2 hours
Definition, separable cost, joint cost, common cost, method of cost allocation–remaining benefits method and alternative justifiable expenditure method.					
Total Lecture hours					36 hours
Text Book(s)					
1.	Engineering Hydrology –K. Subramanya, Tata McGraw- Hill Education.				



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Reference Books

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| 1. | Applied Hydrology -V. T. Chow, McGraw-Hill Science/Engineering/Math; 1 edition (February 1, 1988. |
| 2. | Applied Hydrology –K. N. Mutreja,Tata McGraw-Hill. |
| 3. | Water Resources Engineering-by R.K. Linsley& J.B. Franzini, McGraw-Hill International Book Company. |

Mode of Assessment: Continuous Assessment Test, Quizzes, Assignments, Final Assessment Test

Recommended by Board of Studies

Approved by Academic Council

Date



BCE235PE10T	Plumbing (Water and Sanitation)	L	T	P	C
		2	0	0	2
Pre-requisite: Nil					
Course Objectives:					
1. Study plumbing codes and good engineering practices.					
2. Coordinate plumbing works from inception to completion with Owners, Architects, other consultants and contractors.					
3. Select proper plumbing materials and systems.					
4. Read and interpret plumbing drawings.					
5. Supervise code based plumbing installations.					
6. Understand methods to conserve water and energy.					
7. Protect health and safety of end users.					
8. Enjoy better job opportunities and career options					
Expected Course Outcome:					
Upon completion of this course, the student will be able to					
1. Understand plumbing codes, read and interpret plumbing drawings for good engineering practices.					
2. Apply basic concepts of plumbing in different plumbing works from inception to completion.					
3. Analyze proper plumbing materials and systems for code based plumbing installations.					
4. Evaluate the efficiencies of pumping systems and treatment plants to conserve water and energy and treat water for different uses.					
Module:1 Importance of Codes, Architectural & Structural Coordination and Plumbing Terminologies					6 hours
Scope, purpose; codes and standards in the building industry, UIPC-I, NBC and other codes, Local Municipal Laws, approvals, general regulations					
Provisions for plumbing systems, space planning for plumbing systems. Structural parameters, sunken toilets, location of columns and beams, posttension slabs, importance of ledge walls, water proofing.					
Definitions, use/purpose of- (a) Plumbing fixtures, (b) Traps, (c) Drainage, (d) Water supply.					
Module:2 Plumbing Fixtures and Fittings					6 hours
Definitions of plumbing fixtures, fittings, appliances and appurtenances; maximum flow rates, water closets, bidets, urinals, flushing devices, washbasins, bath/shower, toilets for differently-abled, kitchen sinks, water coolers, drinking fountain, clothes washer, dish washer, mop sink, overflows, strainers, prohibited fixtures, floor drains, floor slopes, location of valves, hot water temperature controls, installation standard dimensions in plan and elevation.					
Module:3 Traps, Interceptors, Indirect Waste and Vents					6 hours
Traps required, trap arms, developed length, trap seals, venting to traps, trap primers, prohibited traps, building traps.					
Discharge for indirect waste piping, nature of contents or systems, proper methods to install indirect waste piping, air gap and air break, sink traps, dish washers, drinking fountains, waste receptors, sterile equipment, appliances, condensers, point of discharge, venting.					
Vent requirement, purpose of venting, trap seal protection, materials, vent connections, flood rim level, termination, vent stacks, water curtain and hydraulic jump, cleanouts, venting of interceptors, introduction to vent sizing.					
Module:4 Sanitary Drainage and Storm Drain					6 hours
Preamble, one pipe and two pipe systems, different pipe materials and jointing methods, hangers and supports, protection of pipes and structures, alternative materials, workmanship, prohibited fittings and practices, hydraulic jump, change in direction of flow, T and Y fittings, cleanouts, pipe grading, fixtures below invert level, suds relief, testing, building sewers, trenching, testing, sumps and pumps, introduction to Drainage Fixture Units (DFU) and sizing of horizontal and vertical pipes.					
Storm drain required, prohibited connections, subsoil drains, sub-drains, gutters, channels or scuppers, roof drains, strainers, leaders, conductors and connections, catchment, collect/capture storm water, discharging storm water, sizing case study as per NBC, safety, traps required, prohibited installations.					
Rain Water Harvesting (RWH) definition, need, catchment, conduits, settlement tanks, treatment, possible uses, recharging pits, NBC requirements, MOEF&CC requirements, and advantages of RWH					
Module:5 Water Supply, Gray and Reclaimed Water					6 hours



Preamble, sources of water, potable and non-potable water, calculating daily water requirement and storage, hot and cold water distribution system, backflow prevention, air gap, cross connection control, pressure and velocity, pipe materials and jointing methods, alternative materials, protection of pipes and structures, pressure controls, types of valves, installation and testing, color codes and arrow marking, introduction to Water Supply Fixture Units (WSFU) and sizing

Hot water systems, individual and centralized systems, solar hot water systems, types, boilers, hot water generators, hot water consumption pattern, introduction to sizing of systems

Definition of gray water, specifications and drawings, safety, total gray water discharge, holding tanks, valves and piping. Reclaimed water systems, pipe identification, installation, safety signs, valves, cross connection, approved uses

Module:6 Pumping Systems and Introduction to WTP & STP	6 hours
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Terminology, pump heads, types of Pumps, applications, pump characteristics, pumps and motors, pump efficiency, motor efficiency, Hydro Pneumatic Systems (HPS), Zoning, Storm Water and Drainage Pumps. Introduction to Net Zero concept, need to reduce and reuse, rating of Water Efficient Plumbing fixtures and fittings, metering and sub-metering, typical daily water and wastewater calculations for a project.

WTP: Sources, utility and treatment of water, parameters of water quality, parts of water treatment plant (WTP), disinfection methods, storage conditions, rainwater harvesting treatment, desalination.

STP: Grey water and black water, characteristics of domestic sewage, sewage treatment methods, aerobic and anaerobic treatment, level of treatment, reclaimed water, and comparison of various methods.

Total Lecture hours	36 hours
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Text Book(s)

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| 1. | A Guide to Good Plumbing Practices- book published by IPA. |
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Reference Books

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| 1. | Uniform Illustrated Plumbing Code-India (UIPC-I) published by IPA and IAPMO (India) |
| 2. | National Building Code (NBC) of India |
| 3. | IS 17650 Part 1 and Part 2 for Water Efficient Plumbing Products |
| | Water Efficient Products-India (WEP-I) published by IPA and IAPMO (India) |
| | Water Efficiency and Sanitation Standard (WE.Stand) published by IPA and IAPMO (India) |
| | Water Pollution, Berry, CBS Publishers |
| | Elements of Water Pollution Control Engineering, O.P. Gupta, Khanna Book Publishing, New Delhi |

Mode of Assessment: Continuous Assessment Test, Quizzes, Assignments, Final Assessment Test

Recommended by Board of Studies

Approved by Academic Council

Date



BCE235PE10P	Plumbing (Water and Sanitation) Laboratory	L	T	P	C
		0	0	2	1
Pre-requisite: Nil					
Course Objectives:					
1. Apply knowledge of plumbing codes and good engineering practices.					
2. Learn plumbing works from inception to completion with Owners, Architects, other consultants and contractors.					
3. Learn selecting proper plumbing materials and systems.					
4. Read and interpret plumbing drawings.					
5. Learn to supervise code based plumbing installations.					
6. Understand methods to conserve water and energy.					
7. Protect health and safety of end users.					
8. Enjoy better job opportunities and career options.					
Expected Course Outcome:					
Upon completion of this course, the student will be able to					
1. Study plumbing codes, read and interpret plumbing drawings for good engineering practices.					
2. Coordinate plumbing works from inception to completion with Owners, Architects, other consultants and contractors.					
3. Select proper plumbing materials and systems for code based plumbing installations.					
4. Understand methods to conserve water and energy.					
List of Practical:					Approx. Hours
1	Attend demonstration by a reputed manufacturer (Min. 2) of water supply pipes and fittings. Cut and joint water supply pipes and fittings as recommended, for the given dimensions.				4
2	Measure the flow of various plumbing fixtures and fittings in residential, commercial and institutional toilets in liters per minute or liters per flush and give your comments.				2
3	Attend demonstration by a reputed manufacturer of DWV pipes and fittings. Cut and joint a trap, drainage pipes and fittings as recommended, for the given dimensions.				3
4	Site visit: Visit any plumbing site and submit a report on observations on plumbing system, architectural and structural provisions, pipe materials, work methods, site conditions, safety and recommendations based on the provisions of UIPC-I/ NBC.				6
	Total				15
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MOPEC-1	ENVIRONMENTAL STUDIES	L	T	P	C
		3	0	0	3
Pre-requisite: Nil					
Course Objectives:					
To create environmental and sustainability awareness among the students. To gain knowledge on different types of pollution in the environment, waste management and Environmental legislation.					
Expected Course Outcome:					
Upon completion of this course, the student will be able to					
1. Describe ecological systems and their interaction with human activities.					
2. Analyze major environmental issues like climate change, pollution, and biodiversity loss.					
3. Evaluate sustainable practices and technologies for environmental conservation.					
4. Understand the role of laws, policies, and ethics in solving environmental problems.					
Module:1 Environmental Science Introduction and Natural Resources					8 hours
Multidisciplinary nature of Environmental Studies: Definition, Scope and Importance – Need for Public Awareness.					
Renewable and non-renewable resources. Natural resources and associated problems. Forest resources, Water resources, Mineral resources, Food resources, Energy resources and Land resources					
Module:2 Ecosystem and Biodiversity					12 hours
Ecosystem: Concept of an ecosystem. Structure and function of an ecosystem. Producers, consumers and decomposers. Energy flow in the ecosystem. Ecological succession. Food chains, food webs and ecological pyramids. Introduction, types, characteristic features, structure and function of the following ecosystem: Forest ecosystem, grassland ecosystem, desert ecosystem, aquatic ecosystems.					
Biodiversity: Introduction. Definition: genetic, species and ecosystem, diversity. Value of biodiversity: consumptive use, productive use, social, ethical, aesthetic and option values. Biodiversity at global, national and local levels. India as a mega diversity nation. Hotspots of biodiversity. Threats to biodiversity: habitat loss, poaching of wildlife, man -wildlife conflicts. Endangered and endemic species of India. Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.					
Module:3 Environmental Pollution,					5 hours
Definition, causes, effects and control measures of air pollution, water pollution, soil pollution, marine pollution, noise pollution, thermal pollution and nuclear hazards, Pollution during Diwali and Fireworks.					
Module:4 Global Environmental Issues and Control Measures					10 hours
Solid waste Management: Causes, effects and control measures of urban and industrial wastes. Role of an individual in prevention of pollution. Case studies.					
Disaster management: Floods, earthquake, cyclone and landslides. E-waste and plastic waste: recycling and reuse					
Water conservation: Rain water harvesting, watershed management. Resettlement and rehabilitation of people: its problems and concerns. Case Studies. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust. Case Studies.					
Module:5 Environmental Ethics					5 hours
Environment Protection Act. -Air (Prevention and Control of Pollution) Act. Water (Prevention and control of Pollution) Act. Wildlife Protection Act. Forest Conservation Act. Issues involved in enforcement of environmental legislation. Public awareness.					
Total Lecture hours					40 hours
Text Book(s)					
1.	Benny Joseph (2005), Environmental Studies, New Delhi, Tata McGraw Hill Publishing Co. Ltd.				
2.	Erach Bharucha (2005), Textbook of Environmental Studies for Undergraduate Courses, Hyderabad, Universities Press.				
Reference Books					
1.	Anji Reddy M. (2007), Textbook of Environmental Sciences and Technology, Hyderabad, BS Publications.				
2.	Y Anjaneyulu (2004), Introduction to Environmental Sciences, BS Publications.				
Mode of Assessment: Continuous Assessment Test, Quizzes, Assignments, Final Assessment Test					



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MNC-AU-05	PROFESSIONAL PRACTICE, LAW & ETHICS		L	T	P	C
			3	0	0	0
Pre-requisite: Nil						
Course Objectives:						
- To familiarize students with the key elements of professional practice, including an introduction to various stakeholders in civil engineering and their respective roles, as well as the fundamental ethical principles governing the profession. - To provide a comprehensive understanding of contracts and contract management in civil engineering, including dispute resolution mechanisms and the legal framework governing labor engagement. - To introduce the concepts of Intellectual Property Rights (IPR) and patents relevant to engineering practice. - To help students understand the diverse roles and responsibilities they are expected to undertake as civil engineering professionals in society. - To develop a solid understanding of the legal and practical dimensions of the civil engineering profession.						
Expected Course Outcome:						
Upon completion of this course, the student will be able to						
1. Understand professional ethics and the roles of various stakeholders in civil engineering practice. 2. Demonstrate knowledge of contracts, contract management, and dispute resolution methods. 3. Explain the basics of Intellectual Property Rights and their relevance to engineering work. 4. Recognize legal and societal responsibilities associated with the civil engineering profession.						
Module:1 Professional Practice and Ethics:						8 hours
Professional Practice and Ethics: Definition of Ethics, Professional Ethics - Engineering Ethics, Personal Ethics; Code of Ethics - Profession, Professionalism, Professional Responsibility, Conflict of Interest, Gift Vs Bribery, Environmental breaches, Negligence, Deficiencies in state-of-the-art; Vigil Mechanism, Whistle blowing, protected disclosures. Introduction to GST- Various Roles of Various Stake holders						
Module:2 Law of Contract:						12 hours
Law of Contract: Nature of Contract and Essential elements of valid contract, Offer and Acceptance, Consideration, Capacity to contract and Free Consent, Legality of Object. Unlawful and illegal agreements, Contingent Contracts, Performance and discharge of Contracts, Remedies for breach of contract. Contracts-II: Indemnity and guarantee, Contract of Agency, Sale of goods Act -1930: General Principles, Conditions & Warranties, Performance of Contract of Sale.						
Module:3 Arbitration, Conciliation and ADR (Alternative Dispute Resolution) system						5 hours
Arbitration, Conciliation and ADR (Alternative Dispute Resolution) system: Arbitration meaning, scope and types distinction between laws of 1940 and 1996; UNCITRAL model law Arbitration and expert determination; Extent of judicial intervention; International commercial arbitration; Arbitration agreements essential and kinds, validity, reference and interim measures by court; Arbitration tribunal appointment, challenge, jurisdiction of arbitral tribunal, powers, grounds of challenge, procedure and court assistance; Distinction between conciliation, negotiation, mediation and arbitration, confidentiality, resort to judicial proceedings, costs; Dispute Resolution Boards; Lok Adalats.						
Module:4 Engagement of Labour and Labour & Other Construction-Related Laws:						10 hours
Role of Labour in Civil Engineering; Methods of engaging labour-on rolls, labour sub-contract, piece rate work; Industrial Disputes Act, 1947; Collective bargaining; Industrial Employment (Standing Orders) Act, 1946; Workmen's Compensation Act, 1923; Building & Other Construction Workers (regulation of employment and conditions of service) Act (1996) and Rules (1998); RERA Act 2017, NBC 2017.						
Module:5 Law relating to Intellectual property:						5 hours
Introduction meaning of intellectual property, main forms of IP, Copyright, Trademarks, Patents and Designs, Secrets; Law relating to Copyright in India including historical evolution of Copy Rights Act, 1957, Meaning of copyright computer programs, Ownership of copyrights and assignment, Criteria of infringement, Piracy in Internet Remedies and procedures in India; Law relating to Patents under Patents Act, 1970						
Total Lecture hours						40 hours
Text Book(s) / Reference Books						
1. Premvir Kapoor, Professional Ethics and Human Values, Khanna Publishing House. 2. B.S. Patil, Legal Aspects of Building and Engineering Contracts, 1974. 3. The National Building Code, BIS, 2017 4. RERA Act, 2017						



5. Meena Rao (2006), Fundamental concepts in Law of Contract, 3rd Edn. Professional Offset
6. Neelima Chandiramani (2000), The Law of Contract: An Outline, 2nd Edn. Avinash Publications Mumbai
7. Avtarsingh (2002), Law of Contract, Eastern Book Co.
8. Dutt (1994), Indian Contract Act, Eastern Law House
9. Anson W.R. (1979), Law of Contract, Oxford University Press
10. Kwatra G.K. (2005), The Arbitration & Conciliation of Law in India with case law on UNCITRAL Model Law on Arbitration, Indian Council of Arbitration
11. Wadhera (2004), Intellectual Property Rights, Universal Law Publishing Co.
12. T. Ramappa (2010), Intellectual Property Rights Law in India, Asia Law House
13. Bare text (2005), Right to Information Act
14. O.P. Malhotra, Law of Industrial Disputes, N.M. Tripathi Publishers
15. K.M. Desai (1946), The Industrial Employment (Standing Orders) Act
16. Rustamji R.F., Introduction to the Law of Industrial Disputes, Asia Publishing House
17. Vee, Charles & Skitmore, Martin (2003) Professional Ethics in the Construction Industry, Engineering Construction and Architectural management, Vol.10, Iss2, pp 117-127, MCB UP Ltd
18. American Society of Civil Engineers (2011) ASCE Code of Ethics – Principles Study and Application
19. Ethics in Engineering- M.W.Martin & R.Schinzinger, McGraw-Hill
20. Engineering Ethics, National Institute for Engineering Ethics, USA
21. www.ieindia.org
22. Engineering ethics: concepts and cases – C. E. Harris, M.S. Pritchard, M.J.Rabins
23. CONSTRUCTION CONTRACTS, <http://www.jnormanstark.com/contract.htm>
24. Internet and Business Handbook, Chap 4, CONTRACTS LAW,
<http://www.laderapress.com/laderapress/contractslaw1.html>
25. Contract & Agreements <http://www.tco.ac.ir/law/English/agreements/General/Contract%20Law/C.htm>
26. Contracts, <http://206.127.69.152/jgretch/crj/211/ch7.ppt>
27. Business & Personal Law. Chapter 7. “ How Contracts Arise”,
<http://yucaipahigh.com/schristensen/lawweb/lawch7.ppt>
28. Types of Contracts, <http://cmsu2.cmsu.edu/public/classes/rahm/meiners.con.ppt>
29. TYPES OF CONTRACTS AND IMPORTANT PROVISIONS,
30. <http://www.worldbank.org/html/opr/consult/guidetxt/types.html>
31. Contract Types/Pricing Arrangements Guideline- 1.4.G (11/04/02),
<http://www.sandia.gov/policy/14g.pdf>

Mode of Assessment: Continuous Assessment Test, Quizzes, Assignments, Final Assessment Test



BCE23502P	ENVIRONMENTAL ENGINEERING LAB	L	T	P	C
		0	0	2	1
Pre-requisite: Nil					
Course Objectives:					
The objective is to develop practical skills in analyzing key physical and chemical parameters of water and wastewater, and to introduce the measurement of ambient noise, fostering an understanding of their significance in environmental quality assessment and management.					
Expected Course Outcome:					
Upon completion of this course, the student will be able to					
1. Accurately measure and interpret physical water quality parameters such as turbidity, electrical conductivity, pH, and various types of solids.					
2. Perform standardized tests for alkalinity, acidity, hardness, Chemical Oxygen Demand (COD), Dissolved Oxygen (DO), and Biochemical Oxygen Demand (BOD), and understand their relevance to water pollution.					
3. Determine the optimum coagulant dose for water treatment, demonstrating an understanding of coagulation-flocculation principles.					
4. Utilize appropriate instrumentation to measure ambient noise levels and interpret the results in the context of environmental standards.					
Test:1 Physical Characterization of water					2 hours
1. Turbidity,					
2. Electrical Conductivity,					
3. pH					
Test:2 Analysis of solids content of water					3 hours
Test:3 Physical Characterization of water					3 hours
1. Alkalinity and acidity,					
2. Hardness,					
3. Chloride					
Test:4 Optimum coagulant dose					2 hours
Test:5 Chemical Oxygen Demand (COD)					2 hours
Test:6 Biochemical Oxygen Demand (BOD)					3 hours
Test:7 Ambient noise measurement					1 hours
Total Lecture hours					16 hours
Mode of Assessment: Continuous Assessment Test, Assignments, Final Assessment Test					